

ISHITA P ACHARYA

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EDUCATION

Mangalore Institute of Technology and Engineering (MITE) Nov 2022 – September 2026 (Expected)

Bachelor of Engineering in Computer Science - (Current GPA 7.92 / 10.0) Moodabidre, Karnataka, India

Relevant Coursework: Data Structures, Algorithms, Operating Systems, Computer Networks, DBMS, Cloud Computing, Parallel Computing, Artificial Intelligence, Machine Learning, Software Engineering, Cryptography and Network Security.

ACADEMIC PROJECTS

Client Pulse-Client Project & Delivery Intelligence Platform

April 2026 - Present

Personal Project

- * Developed and secured RESTful APIs using JWT-based authentication and role-based access control for Admin, Project Manager, Employee, and Client modules, ensuring efficient user management and secure data handling across the platform.
- * Built and optimized automated invoice generation workflows by integrating approved employee timesheets, reducing manual processing effort and improving billing accuracy and operational efficiency.
- * Designed and implemented scalable MongoDB database schemas to support multiple billing models, including Fixed Cost and Retainer-based systems, enabling flexible project and financial management.
- * Improved backend system performance by achieving sub-500ms API response times and maintaining over 70% test coverage using Jest, ensuring reliable, efficient, and well-tested application functionality.

Peripheral Arterial Disease Detection Using Machine Learning

March 2025 - November 2025

Team Project (4)

- * Built an end-to-end ML pipeline to detect Peripheral Arterial Disease (PAD) from unstructured Doppler reports by integrating OCR (Tesseract) with feature extraction and classification models.
- * Developed a Random Forest-based classification system with SMOTE to address class imbalance, with model accuracy of 75% and enhancing recall for minority classes.
- * Built a scalable Flask-based backend pipeline integrating OCR, ML inference, and rule-based validation for automated clinical report processing.
- * Deployed a Flask-based web application for real-time report analysis, enabling automated extraction, prediction, and clinically interpretable results.

ZOE AI Voice Assistant

February 2025

Personal Project

- * Developed an AI-powered voice assistant inspired by JARVIS using Python and Flask.
- * Integrated Speech Recognition to convert voice commands into text for real-time interaction.
- * Used OpenAI API to generate intelligent conversational responses and answer user queries.
- * Implemented ElevenLabs API for realistic text-to-speech output, creating a natural voice interaction experience.

Voice-Controlled Music Player

November 2025 - January 2026

Personal Project

- * Developed a real-time voice-controlled music system enabling hands-free playback control using speech commands.
- * Implemented speech-to-text and text-to-speech pipelines using Speech Recognition for accurate command recognition and natural audio feedback.
- * Integrated Spotify API for remote music control, supporting actions such as play, pause, track navigation, and volume adjustment.
- * Designed a modular backend system using Python and Flask to process voice input, execute commands, and ensure low-latency response.

SKILLS AND CERTIFICATIONS

Programming Languages: C, Python, JavaScript.

Frontend: React.js, HTML5, CSS3, Bootstrap

Backend: Laravel, Node.js, Flask, FastAPI, REST APIs.

AI/ML: Scikit-learn, PyTorch, TensorFlow, Model Evaluation (Precision, Recall, F1-score), RAG, Langchain, OpenCV, Tesseract OCR, Speech Recognition & Voice Processing, Text-to-Speech (TTS), Conversational AI, Image Preprocessing, MediaPipe, NumPy, Pandas,

Databases: MySQL, MongoDB, MongoDB Atlas

Cloud, DevOps & Systems: AWS EC2, S3, Ubuntu, Linux, Git, GitHub, Railway, REST API Design, JWT Authentication.

Frameworks & Tools: Spotify API, ElevenLabs API, OpenAI API, VS Code.

Certifications: Cyber Job Simulation (Delloite), Prompt Wars (MITE), Cybersecurity in the Product Development Lifecycle: From Design to Deployment (MITE), Cybersecurity: Vulnerability Assessment and Exploitation (MITE).